

Personalized Hand-Gesture Recognition

Classical ML Models Compared with Hand-Written Rules

Gesture Presenter — Machine Learning Course Study

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Abstract

Gesture Presenter recognizes hand gestures using MediaPipe landmarks and hand-written geometric rules. This study tested whether classical machine learning could classify one user's gestures more accurately. A decision tree, random forest, and logistic regression model were trained on two recording sessions and tested on a separate third session. Only hand landmarks were stored; no images or video were saved. The random forest performed best with 98.5% trial accuracy, followed by logistic regression with 94.8%, a decision tree with 93.3%, and the existing rules with 79.3%. The main improvement was recognizing natural, non-command movement instead of incorrectly treating it as a command. This is a personalized single-user result and does not show that the models generalize to other people.

1 Goal

Gesture Presenter uses MediaPipe to detect 21 landmarks on a hand. The current application then recognizes gestures with manually selected thresholds, such as finger joint angles and distances between fingertips. These rules are fast and easy to understand, but the same fixed thresholds may not suit every user's hand shape and gesture style.

The research question was:

Can a small, explainable classical ML model recognize one user's static gestures more accurately than the existing hand-written rules across separate recording sessions?

The aim was not to replace MediaPipe. MediaPipe remained responsible for hand detection. The experiment compared two ways of mapping the same landmarks to a gesture:

Camera $\rightarrow$ MediaPipe landmarks $\rightarrow$ {hand-written rules or trained model} $\rightarrow$ gesture

This was designed as a personalized pilot because only one participant and one evening were available. Therefore, the study measures repeatability for one user, not generalization to the public.

2 Data collection

A guided collection script displayed a gesture, a countdown, a 1.5-second recording period, and a rest period. Gesture order was randomized. The participant used the right hand and recorded three sessions:

Nine classes were recorded:

none	point	pinch
pinch_right	open_palm	fist
fist_thumb_up	v_sign	two_finger_scroll

Table 1: Dataset and experimental split.

Session	Frames	Grouped trials	Use
S01	5,284	135	Training
S02	3,509	135	Training
S03	4,762	135	Final evaluation
Total	13,555	405	—

The **none** class contained natural hand movement without an intended command. It is important because false commands are particularly disruptive in a presentation controller.

Camera frames were processed locally and immediately discarded. Each saved CSV row contained the participant, session and trial identifiers, gesture label, timestamp, handedness information, and the (x, y, z) coordinates of all 21 landmarks. No original footage was stored. Landmark data is less sensitive than video, although it should not be described as completely anonymous.

3 Features and models

Each frame was represented by 77 features:

- 63 coordinates: 21 landmarks with x , y , and z values;
- four normalized distances between important hand points;
- horizontal and vertical thumb displacement;
- four finger joint angles; and
- four fingertip-to-wrist distances.

Coordinates were translated so the wrist became the origin and divided by hand size. This reduces sensitivity to where the hand appears in the image and how close it is to the camera.

Three models were trained with balanced class weights and random seed 42:

Decision tree: maximum depth 8 and at least 5 frames per leaf. It was selected first because each prediction can be explained as a short sequence of feature thresholds and compared directly with the manual rules.

Random forest: 250 trees and at least 3 frames per leaf. Combining many trees normally improves stability, but makes individual decisions less direct to explain.

Logistic regression: standardized features and up to 2,000 training iterations. This tests whether simpler, approximately linear boundaries are sufficient for the engineered features.

The existing geometric classifier was the baseline. It uses finger-extension tests, joint angles, fingertip distances, thumb direction, and manually chosen thresholds.

4 Evaluation

All learned models were trained on 8,793 frames from S01 and S02. S03 was excluded from training and used only for evaluation. The comparison program checks model metadata and refuses to evaluate a session that was used for training. All classifiers received the same S03 landmarks in the same order.

Two accuracy measures were calculated:

Frame accuracy scores every camera frame independently.

Trial accuracy combines all frame predictions in one prompted repetition using majority voting.

Trial accuracy is the main measure because adjacent frames are highly similar and are not independent samples. One complete gesture repetition is a more meaningful unit than one frame.

For each gesture, precision and recall describe different types of error:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN},$$

where TP is the number of correct predictions of that gesture, FP is the number of other gestures incorrectly predicted as that gesture, and FN is the number of real examples of that gesture that the classifier missed. Precision therefore asks, “When the model predicts this gesture, how often is it correct?” Recall asks, “Of all real examples of this gesture, how many did the model find?”

The F1 score combines precision and recall using their harmonic mean:

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.$$

F1 ranges from 0 to 1. A value of 1 means that a gesture was found every time and was never predicted incorrectly. A low value means that precision, recall, or both were poor. F1 is useful here because accuracy alone can hide a serious problem with one gesture class. For example, a classifier may be accurate overall while frequently missing pinches or treating natural movement as a command.

Macro F1 is the unweighted mean of the F1 scores for all K gesture classes:

$$\text{Macro F1} = \frac{1}{K} \sum_{k=1}^K F1_k.$$

Every gesture contributes equally to macro F1, regardless of how many examples it has. This makes it appropriate for this dataset because the held-out class supports are not exactly equal.

This comparison covers static classification only. The live application also uses hold durations, debouncing, swipe tracking, scrolling trajectories, and cooldowns. Therefore, the experiment does not yet measure actual unintended clicks or slide changes.

5 Results

5.1 Overall comparison

Table 2: Results on the held-out S03 session.

Classifier	Frame accuracy	Trial accuracy	Correct trials
Random forest	98.8%	98.5%	133/135
Logistic regression	95.7%	94.8%	128/135
Decision tree	91.7%	93.3%	126/135
Hand-written rules	80.0%	79.3%	107/135

All three learned models outperformed the rules. The random forest improved trial accuracy by 19.2 percentage points and made only two trial-level errors. The single decision tree was less accurate than the forest but remains much easier to explain because it uses one visible path of threshold decisions.

5.2 Performance by gesture

Table 3 summarizes trial-level F1 scores. F1 combines precision (how often a prediction is correct) and recall (how many real examples are found).

Table 3: Trial-level F1 score by gesture.

Gesture	Forest	Logistic	Tree	Rules
fist	1.00	1.00	0.97	0.73
fist_thumb_up	1.00	0.89	0.96	0.89
none	1.00	0.93	0.90	0.00
open_palm	1.00	1.00	1.00	0.84
pinch	0.97	0.88	0.77	0.83
pinch_right	0.97	0.93	0.90	0.85
point	0.96	0.96	0.96	0.63
two_finger_scroll	0.97	0.94	0.97	1.00
v_sign	1.00	1.00	0.96	0.92
Macro average	0.99	0.95	0.93	0.67

The most important result was the **none** class:

- the forest and decision tree recognized all 14 **none** trials;
- logistic regression recognized 13 of 14;
- the rules recognized 0 of 14.

The rules interpreted the 14 natural-movement trials as seven open palms, three fists, two pinches, one point, and one thumb-left gesture. These would be potential false commands before temporal filtering is applied.

The random forest made only two errors: one true index pinch was predicted as point, and one two-finger-scroll pose was predicted as right-click pinch. The single decision tree’s main weakness was index pinch. It recognized 12 of 19 pinch trials; the other seven were distributed across five other classes. The forest increased pinch recall from 0.63 to 0.95, showing the benefit of combining multiple trees.

Logistic regression produced strong results with a simpler decision function. Its mistakes were mainly pinch and two-finger-scroll examples. The numerical backend printed overflow warnings during fitting, but the input features, trained coefficients, predictions, and model output were checked and were all finite. The training completed in 78 iterations. This score should still be reproduced in a clean environment before being treated as a final benchmark.

6 What the results mean

For this participant, learned decision boundaries transferred from two sessions to a separate third session better than the fixed rules. The random forest was the best predictive model. The decision tree remains the best choice for direct XAI because its learned thresholds can be read and compared with the existing manual thresholds.

The strongest product-relevant finding is not only higher accuracy. It is the ability to reject natural hand movement. The current rules are good at several clearly posed gestures, including two-finger scroll, but often force ordinary movement into a command class. The learned models captured the participant’s non-command poses much better.

The valid conclusion is:

For one participant, three classical ML models trained on two sessions achieved higher static gesture-classification accuracy than the existing rules on a held-out third session. The random forest was most accurate, while the single decision tree offered the clearest explanations.

It would not be valid to claim that these models work for arbitrary users or that the random forest is already better in the complete live application.

7 Limitations and next steps

1. Only one person participated, so there is no cross-user evidence.
2. Adjacent frames are correlated. Trial-level results are more meaningful than the larger frame count.
3. The class supports in S03 range from 11 to 19 instead of exactly 15. This is consistent with restarting randomized collection using the same session and trial identifiers, which can mix frames inside a grouped trial. A final confirmation session should use a fresh session ID and must not be used to tune the existing models.
4. The models were compared using fixed configurations. S03 must not now be reused for hyperparameter selection, because that would leak test information.
5. The next XAI step is to visualize the decision tree, inspect feature importance, and explain misclassified pinch trials. The next product-level experiment should apply temporal filtering and measure false actions per minute.

A Reproduction

The model type can be `tree`, `forest`, or `logistic`:

```
venv/bin/python -m research.train \
  research/data/landmarks.csv \
  --model forest \
  --exclude-session S03 \
  --output research/models/forest.joblib

venv/bin/python -m research.compare \
  --model research/models/forest.joblib \
  --session S03 \
  research/data/landmarks.csv
```